Convection in a Pan

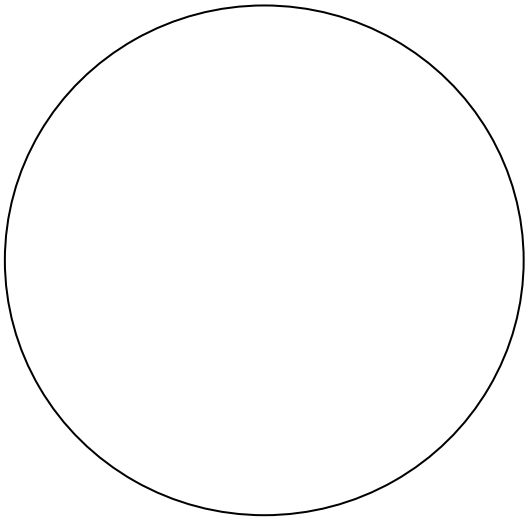
After you light the candle under the pie pan in front of you, wait 3 minutes and carefully observe the surface of the soapy water.

With your group, try these things to help you figure out what is happening:

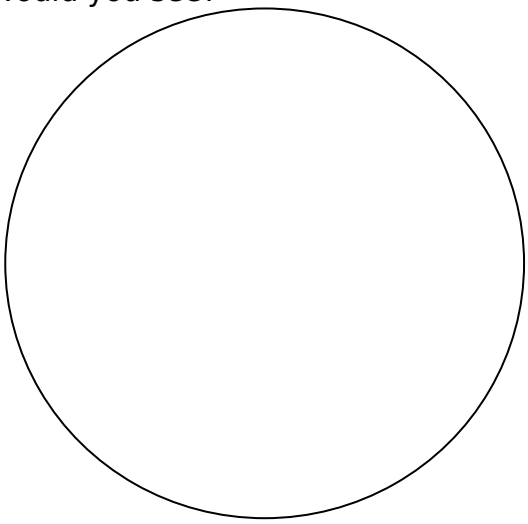
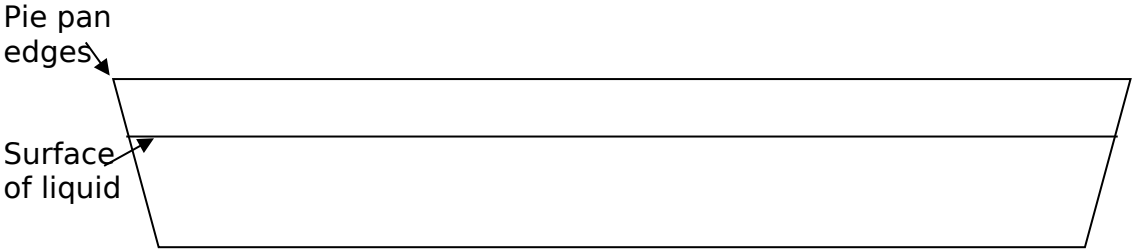
1. Add one drop of food coloring to the center of the liquid, directly above the candle flame.
2. Add one drop of food coloring of a different color to the liquid between the center and the rim of the pan.
3. Add one drop of food coloring of a different color to the edge of the liquid, near the rim of the pan.

When you think you know how the liquid is moving, use arrows to show the direction of motion of the liquid from these 3 different views:

What do you see looking down onto the surface of the liquid?



If the pan were made of glass and if you could lie down on the table and watch the liquid from below, what would you see?



What would you see if you had x-ray vision and could look through the pan from the side?